

Quadratics & Polynomials



Factoring quadratics

- 1 Solve $x^2 + 7x + 12 = 0$ by factoring.
 - 2 Solve $x^2 - 9x + 20 = 0$ by factoring.
 - 3 Solve $2x^2 + 5x - 3 = 0$ by factoring.
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Vertex form and completing the square

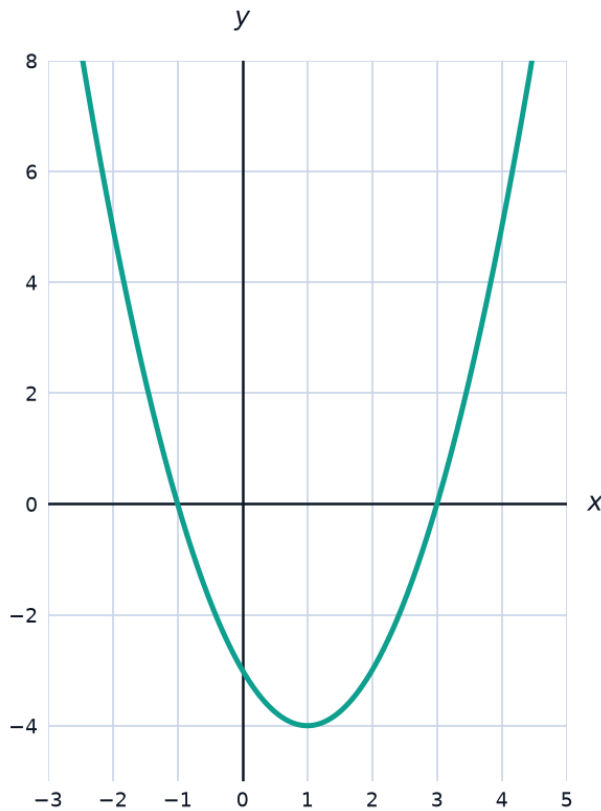
- 4 Write $y = x^2 - 6x + 5$ in vertex form by completing the square, and state the vertex.
 - 5 Solve $x^2 + 8x + 3 = 0$ by completing the square.
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The quadratic formula and discriminant

- 6 Solve $3x^2 - 5x - 2 = 0$ using the quadratic formula.
 - 7 Find the discriminant of $2x^2 + 3x + 5 = 0$ and state the nature of its roots.
 - 8 For what values of k does $x^2 + kx + 16 = 0$ have exactly one repeated real root?
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Graphing parabolas

- 9 The graph of $y = x^2 - 2x - 3$ is shown below. Use the graph to state the x -intercepts, then confirm your answer algebraically by factoring.



- 10 The graph of $y = -2x^2 + 8x - 3$ opens which direction, and what is the x -coordinate of its vertex?

Polynomial multiplication and special products

- 11 Expand $(2x - 5)^2$.
- 12 Multiply $(x + 7)(x - 7)$.
- 13 Multiply $(3x + 4)(3x - 4)$.

Polynomial division and the remainder theorem

- 14 Use synthetic division to divide $x^3 - 2x^2 - 5x + 6$ by $(x - 1)$, and state the quotient and remainder.
- 15 Use the Remainder Theorem to find the remainder when $x^3 + 4x^2 - x + 2$ is divided by $(x + 2)$.

Quadratic word problems

- 16 A ball is thrown upward so that its height is modeled by $h(t) = -16t^2 + 64t + 5$ (in feet), where t is time in seconds. Find the time at which the ball reaches its maximum height, and find that maximum height.
- 17 The area of a rectangle is $x^2 + 2x - 15$ square meters, and its width is $x - 3$. Find its length in terms of x by factoring.
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Sum and product of roots

- 18 For $x^2 - 7x + 10 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.
- 19 A quadratic equation has roots 3 and -5 . Write the equation in standard form.
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A well-designed word problem: maximizing weekly profit

- 20 This question has three parts. A company's weekly profit (in thousands of dollars) from producing x hundred units is modeled by $P(x) = -2x^2 + 20x - 32$. (a) Factor $P(x)$ to find the break-even production levels, where profit is zero. (b) Find the vertex of the parabola to determine the production level that maximizes profit, and state the maximum profit. (c) State the range of production levels (in hundreds of units) for which profit is positive.
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Multiple choice — easy

- 21 Solve $x^2 - 15x + 54 = 0$ by factoring.
- a. 6, 8
 - b. 7, 9
 - c. 6, 9
 - d. $-6, -9$

22 Solve $x^2 - 6x = 0$ by factoring.

a. 1, 6

b. 0, 5

c. 0, - 6

d. 0, 6

23 Solve $x^2 + 5x + 4 = 0$ by factoring.

a. 4, 1

b. -4, - 2

c. -3, - 1

d. -4, - 1

24 Solve $x^2 - 9 = 0$ by factoring.

a. -3, 3

b. 3, - 3

c. -3, 2

d. -2, 3

25 Solve $x^2 + 4x = 0$ by factoring.

a. 4, 0

b. -3, 0

c. -4, 0

d. -4, - 1

- 26 Solve $x^2 - 2x = 0$ by factoring.
- a. 0, 2
 - b. 0, -2
 - c. 0, 1
 - d. 1, 2
- 27 Solve $x^2 + 3x + 2 = 0$ by factoring.
- a. 2, 1
 - b. -1, -1
 - c. -2, -1
 - d. -2, -2
- 28 Solve $x^2 - 2x - 63 = 0$ by factoring.
- a. -6, 9
 - b. -7, 8
 - c. 7, -9
 - d. -7, 9
- 29 Solve $x^2 - 4x - 21 = 0$ by factoring.
- a. -2, 7
 - b. -3, 7
 - c. 3, -7
 - d. -3, 6

30 Solve $x^2 - x - 12 = 0$ by factoring.

a. $-3, 4$

b. $-2, 4$

c. $-3, 3$

d. $3, -4$

31 Solve $x^2 - 13x + 42 = 0$ by factoring.

a. $6, 6$

b. $6, 7$

c. $7, 7$

d. $-6, -7$

32 Solve $x^2 + 11x + 28 = 0$ by factoring.

a. $-7, -4$

b. $-6, -4$

c. $-7, -5$

d. $7, 4$

33 Solve $x^2 + 12x + 35 = 0$ by factoring.

a. $-6, -5$

b. $7, 5$

c. $-7, -5$

d. $-7, -6$

- 34 Solve $x^2 - 9x = 0$ by factoring.
- a. 0, 8
 - b. 0, -9
 - c. 0, 9
 - d. 1, 9
- 35 Solve $x^2 - 5x - 14 = 0$ by factoring.
- a. -2, 7
 - b. -2, 6
 - c. 2, -7
 - d. -1, 7
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Multiple choice — medium

- 36 Solve $2x^2 - 17x + 30 = 0$ using the quadratic formula.
- a. $\frac{5}{2}, 5$
 - b. $\frac{7}{2}, 6$
 - c. $\frac{5}{2}, 6$
 - d. $-\frac{5}{2}, -6$
- 37 Solve $2x^2 + 8x - 24 = 0$ using the quadratic formula.
- a. -6, 1
 - b. 6, -2
 - c. -5, 2
 - d. -6, 2

38 Solve $3x^2 - 9x - 30 = 0$ using the quadratic formula.

a. $-2, 5$

b. $-1, 5$

c. $-2, 4$

d. $2, -5$

39 Solve $x^2 - 8x + 12 = 0$ using the quadratic formula.

a. $-2, -6$

b. $3, 6$

c. $2, 5$

d. $2, 6$

40 Solve $2x^2 - 20x + 48 = 0$ using the quadratic formula.

a. $4, 5$

b. $4, 6$

c. $5, 6$

d. $-4, -6$

41 Solve $2x^2 - 6x - 8 = 0$ using the quadratic formula.

a. $-1, 4$

b. $-1, 3$

c. $0, 4$

d. $1, -4$

42 Solve $3x^2 + 3x - 60 = 0$ using the quadratic formula.

a. 5, -4

b. -5, 4

c. -5, 3

d. -4, 4

43 Solve $x^2 - 3x - 10 = 0$ using the quadratic formula.

a. -2, 5

b. -2, 4

c. -1, 5

d. 2, -5

44 Solve $2x^2 - 12x = 0$ using the quadratic formula.

a. 0, -6

b. 0, 5

c. 1, 6

d. 0, 6

45 Solve $2x^2 + 6x - 8 = 0$ using the quadratic formula.

a. -4, 0

b. -4, 1

c. 4, -1

d. -3, 1

- 46 Solve $2x^2 - 6x = 0$ using the quadratic formula.
- a. 0, 2
 - b. 0, 3
 - c. 0, -3
 - d. 1, 3
- 47 Solve $x^2 - x - 2 = 0$ using the quadratic formula.
- a. -1, 1
 - b. -1, 2
 - c. 1, -2
 - d. 0, 2
- 48 Solve $3x^2 + 12x + 9 = 0$ using the quadratic formula.
- a. -3, -1
 - b. -2, -1
 - c. -3, -2
 - d. 3, 1
- 49 Solve $x^2 - x - 30 = 0$ using the quadratic formula.
- a. -5, 5
 - b. 5, -6
 - c. -4, 6
 - d. -5, 6

50 Solve $2x^2 - 14x + 12 = 0$ using the quadratic formula.

- a. $-1, -6$
- b. $2, 6$
- c. $1, 6$
- d. $1, 5$

Multiple choice — hard

51 Find the vertex of the parabola $y = -2x^2 - 28x - 92$.

- a. $(-7, 6)$
- b. $(-5, 6)$
- c. $(7, 6)$
- d. $(-7, -6)$

52 Find the vertex of the parabola $y = -x^2 + 16x - 71$.

- a. $(10, -7)$
- b. $(8, -7)$
- c. $(8, 7)$
- d. $(-8, -7)$

53 Find the vertex of the parabola $y = 3x^2 - 6x + 18$.

- a. $(1, 15)$
- b. $(-1, 15)$
- c. $(3, 15)$
- d. $(1, -15)$

54 Find the vertex of the parabola $y = 3x^2 - 42x + 147$.

- a. $(-7, 0)$
- b. $(9, 0)$
- c. $(7, 2)$
- d. $(7, 0)$

55 Find the vertex of the parabola $y = -3x^2 - 24x - 47$.

- a. $(-2, 1)$
- b. $(-4, 1)$
- c. $(-4, -1)$
- d. $(4, 1)$

56 Multiply $(5x + 7)^2$.

- a. $25x^2 + 70x - 49$
- b. $25x^2 + 70x + 49$
- c. $25x^2 + 49$
- d. $25x^2 - 70x + 49$

57 Multiply $(2x + 11)^2$.

- a. $4x^2 - 44x + 121$
- b. $4x^2 + 44x - 121$
- c. $4x^2 + 44x + 121$
- d. $4x^2 + 121$

58 Multiply $(5x + 6)(5x - 6)$.

a. $25x^2 + 36$

b. $25x^2 + 60x - 36$

c. $25x^2 - 38$

d. $25x^2 - 36$

59 Multiply $(4x + 9)^2$.

a. $16x^2 + 81$

b. $16x^2 + 72x + 81$

c. $16x^2 + 72x - 81$

d. $16x^2 - 72x + 81$

60 Multiply $(2x + 9)(2x - 9)$.

a. $4x^2 + 81$

b. $4x^2 - 81$

c. $4x^2 + 36x - 81$

d. $4x^2 - 83$

61 For $2x^2 + 12x + 16 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.

a. $-6, 8$

b. $8, -6$

c. $6, 8$

d. $-6, -8$

- 62 For $2x^2 - 13x + 8 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.
- a. $\frac{13}{2}, -4$
 - b. $-\frac{13}{2}, 4$
 - c. $\frac{13}{2}, 4$
 - d. $4, \frac{13}{2}$
- 63 For $2x^2 + 14x + 2 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.
- a. $1, -7$
 - b. $-7, 1$
 - c. $7, 1$
 - d. $-7, -1$
- 64 For $x^2 + 6x + 6 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.
- a. $6, -6$
 - b. $6, 6$
 - c. $-6, 6$
 - d. $-6, -6$
- 65 For $x^2 + 13x + 7 = 0$, find the sum and product of its roots without solving, using $\text{sum} = -\frac{b}{a}$ and $\text{product} = \frac{c}{a}$.
- a. $-13, -7$
 - b. $-13, 7$
 - c. $13, 7$
 - d. $7, -13$